

AMENDMENT NO. 2 NOVEMBER 2020

TO

**IS 1893 (PART 1) : 2016 CRITERIA FOR EARTHQUAKE
RESISTANT DESIGN OF STRUCTURES**

PART 1 GENERAL PROVISIONS AND BUILDINGS

(Sixth Revision)

(Page 5, clause 4.25) — Insert the following new definition at the end:

‘4.26 Floating Column — A vertical element (column) which at its lower level (termination level) rests on a beam.’

(Page 8, clause 6.3.3.1) — Substitute the following for the existing:

‘6.3.3.1 Effects due to vertical earthquake shaking shall be considered,

- a) for the stability of embankments, bridges and dams; and
- b) for the safety of buildings, embankments, bridges and dams or their components, including in the following cases:
 - 1) Structure located in Seismic Zone IV or V,
 - 2) Structure has vertical or plan irregularities,
 - 3) Structure is rested on soft soil,
 - 4) Structures with long spans, which cause amplification of oscillations due to their flexibility,
 - 5) Structure has large overhangs of structural members or sub-systems,
 - 6) Prestressed structures, and
 - 7) Structures with prestressed members.’

(Page 8, clause 6.3.4) — Substitute the following for the existing:

‘6.3.4 *Combinations to Account for Two or Three Directional Earthquake Ground Shaking*’

Amendment No. 2 to IS 1893 (Part 1) : 2016

[Page 14, Table 5, Sl No. (i), last para] — Substitute the following for the existing:

‘In a building with torsional irregularity, when the horizontal total displacement $\Delta_{\max}$ (which is larger in amplitude) at one end of the building in plan is:

- i) in the range of $1.2 \Delta_{\text{ave}}$ to $1.4 \Delta_{\text{ave}}$ (where $\Delta_{\text{ave}} = (\Delta_{\max} + \Delta_{\min})/2$, in which $\Delta_{\min}$ is the horizontal total displacement, which is smaller in amplitude at the other end), (a) the building configuration shall be revised to ensure that the natural period of the fundamental torsional mode of oscillation shall be smaller than those of the first two translational modes along each of the principal plan directions, and then (b) three dimensional dynamic analysis method shall be adopted.*
- ii) more than $1.4 \Delta_{\text{ave}}$, the structural configuration of the building shall be revised.’*

[Page 16, Table 5, Sl No. (ii), last para] — Substitute the following for the existing:

‘In a building with re-entrant corners, three-dimensional dynamic analysis method with flexible floor diaphragm shall be adopted to capture the concentration of forces generated in the re-entrant corners especially in the floor diaphragm and special elements adjoining the re-entrant corner. This is in addition to the case of rigid diaphragm analysis, if applicable, and the worst effect considered.’

[Page 16, Table 5, Sl No. (iii), last para] — Substitute the following for the existing:

‘In a building with floor slabs having excessive cut-outs or openings, if the area of the geometric cut-out is,

- a) less than or equal to 50 percent, the floor slab shall be taken as rigid or flexible in plane in structural analysis of the building depending on the location of and size of openings.*
- b) more than 50 percent, the floor slab shall be taken as flexible in plane in structural analysis of the building’.*

[Page 16, Table 5, Sl No.(iv), para (b)] — Substitute the following for the existing:

‘b) the following two conditions shall be satisfied, if the building is located in Seismic Zones III, IV and V:

- 1) The forces and moments due to earthquake effects in the elements connecting the two vertical elements with out of plane offset elements,*

Amendment No. 2 to IS 1893 (Part 1) : 2016

the vertical element supporting the offset, and connections shall be enhanced by a factor of at least 2.5.

- 2) *Lateral drift shall be less than 0.2 percent in the storey having the offset and also in the storeys below.'*

[Page 16, Table 5, Sl No. (v), last para] — Substitute the following for the existing:

'In a building with non-parallel lateral force resisting system, the building shall be analyzed for load combinations mentioned in 6.3.2.2 or 6.3.4.1.'

[Page 16, Table 6, Sl No. (i)] — Substitute the following for the existing:

'i) Stiffness irregularity (Soft Storey)

A soft storey is a storey whose lateral stiffness is less than that of the storey above.

In a building with stiffness irregularity, the following shall be complied with:

- 1) *Dynamic Analysis shall be employed to capture the actual distribution of lateral stiffness along the height of the building; and*
- 2) *The inter-storey drift shall be limited to 0.2 percent in that storey and all storeys below, if any, with stiffness irregularity.*

Also, in buildings with URM infills, when contribution of masonry infills exceeds 20 percent in the structural plan density (SPD) of the building, the effect of URM infills shall be considered by explicitly modelling the same in structural analysis. In buildings with stiffness irregularity arising out of URM Infills, provisions of 7.9 shall apply.

NOTE — This provision is not applicable to such storeys, which have: (a) lower height, where services and utilities are housed, or (b) outrigger frame members placed in them.'

[Page 16, Table 6, Sl No. (ii), last para] — Substitute the following for the existing:

'In a building with mass irregularity and located in Seismic Zones III, IV and V, the earthquake effects shall be estimated by dynamic analysis (as per 7.7).'

[Page 16, Table 6, Sl No. (iii), last para] — Substitute the following for the existing:

'In a building with vertical geometric irregularity and located in Seismic Zones III, IV and V, the earthquake effects shall be estimated by Dynamic Analysis (as per 7.7).'

Amendment No. 2 to IS 1893 (Part 1) : 2016

[Page 16, Table 6, Sl No. (iv), last para] — Substitute the following for the existing:

'In a building with in-plane discontinuity and located in Seismic Zone II, the lateral drift of the building under the design lateral force shall be limited to 0.2 percent of the building height. Buildings with in-plane discontinuity shall not be permitted in Seismic Zones III, IV and V.'

[Page 16, Table 6, Sl No. (v), last para] — Substitute the following for the existing:

'Buildings with strength irregularity shall not be permitted. In case, the weak storey is because of URM infills, provisions of 7.10 shall be followed.'

[Page 16, Table 6, Sl No. (vi)] — Substitute the following for the existing:

‘vi) Floating Columns

Such columns are likely to cause concentrated damage in the structure, and are undesirable.

A building with floating columns shall not be permitted, if the floating columns are part of or supporting the primary lateral load resisting system.'

[Page 16, Table 6, Sl No. (vii)] — Substitute the following for the existing:

‘vii) Irregular Modes of Oscillation in Two Principal Plan Directions

Stiffnesses of beams, columns, braces and structural walls determine the lateral stiffness of a building in each principal plan direction. A building is said to have lateral irregularity in a principal plan direction, if:

- a) the first three lateral translational modes contribute less than 65 percent mass participation factor in each principal plan direction, and
- b) the fundamental lateral translational natural periods of the building in the two principal plan directions are closer to each other by 10 percent of the larger value.

In a building with lateral irregularity in any of its two principal plan directions, and which is located in:

- 1) *Seismic Zones II and III, it shall be ensured that the first three lateral translational modes together contribute at least 65 percent mass participation factor in each principal plan direction.*

Amendment No. 2 to IS 1893 (Part 1) : 2016

2) *Seismic Zones IV and V, it shall be ensured that:*

- a) *the first three lateral translational modes together contribute at least 65 percent mass participation factor in each principal plan direction, and*
- b) *the fundamental lateral translational natural periods of the building in the two principal plan directions are away from each other by at least 10 percent of the larger value.*

NOTE — This provision is not applicable to buildings with large podiums in the lower storeys.’

(Page 17, Fig. 4A, middle figure) — Delete.

(Page 18, Fig. 4E, middle figure) — Delete.

[Page 19, Table 8, Sl No. (i), col. 2, line 3] — Substitute ‘educational buildings’ for ‘schools’.

[Page 20, Table 9, Sl No. (ii)(a), line 2] — Insert the words ‘(see Note 1)’ at the end.

[Page 20, Table 9, Sl No. (iv), title] — Delete the words ‘(see Note 3)’.

[Page 20, Table 9, Sl No. (v)] — Substitute the following for the existing:

‘v) **Flat Slab — Structural Wall Systems** (see Note 4)

RC building with the features given below:

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- a) Ductile RC structural walls (which are designed to resist 100 percent of the design lateral force),
- b) Perimeter RC SMRFs (which are designed to independently resist 25 percent of the design lateral force), and
- c) Preferably a system (such as outrigger belt truss system) connecting the core ductile RC structural walls and the perimeter RC SMRFs.’

[Page 20, Table 9, Note 1, line 1] — Substitute ‘Structures’ for ‘RC and steel structures’.

Amendment No. 2 to IS 1893 (Part 1) : 2016

(Page 20, Table 9, Note 3) — Substitute the following for the existing note:
‘See 7.2.7’.

(Page 20, Table 9, Note 4) — Substitute the following for the existing:

‘4 In these buildings, (a) punching shear failure shall be avoided, and (b) lateral drift at the roof under lateral force shall not exceed 0.1 percent. Lateral drift of the flat slab buildings shall be estimated: (i) considering total lateral displacement, including torsional effects, and (ii) using three-dimensional models. Scaling need not be done for displacement response quantities, as stated in 7.7.3.2.’

[Page 21, clause 7.6.2 (b) (see also Amendment No. 1)] — Substitute the following for the existing expression of A_w :

$$A_w = \sum_{i=1}^{N_w} \left[A_{wi} \left\{ 0.2 + \left(\frac{L_{wi}}{h} \right)^2 \right\} \right]$$

[Page 21, clause 7.6.2(c)] — Insert the following at the end of (c):

‘7.6.2.1 The value of T_a obtained shall neither be taken to be more than that given in 7.6.2 (a) nor less than that given in 7.6.2 (c).’

(Page 23, clause 7.7.3, para 2, line 1) — Substitute the following for the existing:

‘7.7.3.1 When $V_B < \overline{V_B}$, the force response quantities.....’

(Page 23, clause 7.7.3, para 2, last line) — Insert the following new clause at the end:

‘7.7.3.2 When $V_B < \overline{V_B}$, the displacement response quantities need not be scaled.’

(Page 25, clause 7.9.1, last line) — Insert the following new clause at the end:

‘7.9.1.1 RC members shall be designed for the governing combinations of stress resultants arising from structural analysis of:

- a) Bare frame building, and
- b) Frame building with URM infills.’

Amendment No. 2 to IS 1893 (Part 1) : 2016

(Page 26, clause 7.11.1.2) — Substitute the following for the existing:

‘7.11.1.2 Displacement estimates obtained from dynamic analysis need not be scaled, as stated in 7.7.3.2.’

(Page 27, clause 7.11.3, line 5) — Substitute ‘for the two’ for ‘of the two’.

(Page 39, Fig. 8) — Substitute ‘ $(N_1)_{60CS}$ ’ for ‘ $(N_1)_{60}$ ’.

(Page 39, Fig. 8) — Substitute ‘CYCLIC RESISTANCE RATIO ($CRR_{7.5}$)’ in place of the existing ‘CYCLIC STRESS RATIO (CSR) OR CYCLIC RESISTANCE RATIO (CRR)’.

(Page 40, Fig. 9) — Substitute ‘CYCLIC RESISTANCE RATIO ($CRR_{7.5}$)’ in place of the existing ‘CYCLIC STRESS RATIO (CSR) OR CYCLIC RESISTANCE RATIO (CRR)’.

(Page 41, Fig. 10) — Substitute ‘CYCLIC RESISTANCE RATIO ($CRR_{7.5}$)’ in place of the existing ‘CYCLIC STRESS OR RESISTANCE RATIO. CSR OR CRR’.